

Fire Fighting System and Obstacle Avoiding Robot

CSE 438 — Robotics Lab | University of Asia Pacific

Department of Computer Science and Engineering, Dhaka, Bangladesh

Md. Shahriar Hossain (22201196) — 22201196@uap-bd.edu

Shihad Md. Akash (22201203) — 22201203@uap-bd.edu

Abdullah al Masud (22201200) — 22201200@uap-bd.edu

Roudro Saha (22201205) — 22201205@uap-bd.edu

Abstract — This paper presents the design and implementation of an autonomous fire-fighting robot with integrated obstacle avoidance capability. The system employs an HC-SR04 ultrasonic sensor for real-time obstacle detection and dual infrared (IR) flame sensors for fire identification, orchestrated by an Arduino UNO (ATmega328P) microcontroller. A three-mode prioritized finite-state machine (FSM) — Fire Mode, Obstacle Mode, and Normal Mode — governs all robot behaviour, with RGB LED visual feedback for each operational state. Built on a 4-Wheel Drive (4WD) acrylic chassis with an L298N motor driver and a relay-controlled submersible water pump, the robot detects obstacles within 20 cm and activates the pump upon flame detection within approximately 25 cm. A 5-sample analog averaging filter eliminates false flame-detection triggers from ambient infrared noise. The complete active system was realized at a total hardware cost of approximately BDT 3,800, demonstrating that reliable embedded robotic safety systems are achievable on a low-cost platform suitable for wide deployment in Bangladesh.

Index Terms — Fire Fighting Robot, Obstacle Avoidance, Arduino UNO, Embedded Systems, HC-SR04, IR Flame Sensor, L298N Motor Driver, Finite-State Machine, Sensor Fusion.

I. INTRODUCTION

Fire accidents and physical collisions pose serious risks in industrial, residential, and manufacturing environments. In Bangladesh, garment factories and manufacturing plants are particularly susceptible to fire hazards, historically resulting in large-scale casualties and property losses. Deploying autonomous robotic systems capable of detecting and responding to such hazards without direct human intervention represents a practical and scalable safety solution.

This project, developed under CSE 438 — Robotics Lab at the University of Asia Pacific, presents a Fire Fighting System and Obstacle Avoiding Robot. The robot combines embedded control logic, multi-sensor interfacing, and real-time actuation to suppress fires and navigate independently around obstacles. The system is built on an Arduino UNO microcontroller and employs a three-mode prioritized control strategy: Fire Mode (highest priority), Obstacle Mode, and Normal Navigation Mode, with RGB LED indicators providing real-time visual state feedback.

A pre-project market survey conducted in Dhaka identified locally available, cost-effective components — particularly IR flame sensors at BDT 150 each — suitable for high-heat environments. A complementary impact assessment established garment factories and manufacturing plants as the primary target deployment zones, where this robot can reduce human exposure during early-stage fire events. The total active bill of materials was validated at approximately BDT 3,800.

The objectives of this project are to: (1) design a fire-fighting embedded robotic platform that navigates autonomously in an unstructured indoor environment; (2) detect and avoid physical obstacles in real time using ultrasonic distance measurement; and (3) identify and suppress fire sources through an automated water-spray mechanism activated by dual IR flame sensors.

II. RELATED WORKS

The development of autonomous robots for fire-fighting and obstacle avoidance applications has been extensively studied. This section reviews the key works that informed the design of the proposed system.

Pathak et al. [1] demonstrated an ultrasonic obstacle-avoiding robot that successfully navigates around static and dynamic obstacles by measuring distance and altering direction upon threshold violation. Hossain and Rahman [2] implemented an Arduino-based mobile robot combining ultrasonic sensing with PWM motor control for autonomous navigation, directly validating the sensing and actuation strategy adopted here. Kumar et al. [3] further confirmed the HC-SR04 ultrasonic sensor's ± 3 mm accuracy at the 20 cm operating distance used in this project.

For fire detection and suppression, Tiwari and Kumar [4] proposed a sensor-controlled firefighting robot using infrared sensors to identify flames and activate a water pump, establishing that IR-based flame detection is reliable and cost-effective on embedded platforms. Kumbhar and Patil [5] elaborated on modular hardware design for scalable embedded robotics, which informed the component-level and power-isolation architecture of the proposed system. Mishra [6] further developed a mobile firefighting robot demonstrating effective small-scale fire suppression, and Kulkarni and Sali [7] reinforced the feasibility of sensor-based Arduino automation systems in real-time control — a key validation of the microcontroller-centric design adopted here.

Mendoza [8] showed that fusing ultrasonic and infrared sensor data substantially improves robot accuracy and responsiveness, providing the multi-sensor rationale for this project. The theoretical foundations for sensing, actuation, and control draw from the seminal robotics texts of Dudek and

Jenkin [9], Siciliano and Khatib [10], and Craig [11], which together underpin the kinematics, control architecture, and embedded system design principles applied throughout. Industrial deployment strategy was informed by Saha [12], whose work on autonomous mobile robotics for industrial applications contextualizes the proposed robot's safety-focused use case.

More recent contributions have explored household and IoT-connected safety systems. Islam and Islam [13] designed a compact household fire suppression robot, demonstrating embedded safety at residential scale. Singh [14] proposed an IoT-based fire detection and control system enabling remote monitoring — a planned Phase 2 enhancement for the proposed system. Collectively, these works validate every major design decision in this project and establish a strong foundation for the fire-fighting and obstacle avoidance integration achieved here.

III. ROBOT STRUCTURE DETAILS / HARDWARE SETUP

A. System Architecture

The embedded system is designed as a modular architecture integrating sensing, processing, actuation, fire-extinguishing, and power management subsystems. Fig. 1 shows the complete hardware architecture block diagram with all input sensors, the central processing unit, and all output actuators with their respective pin assignments.

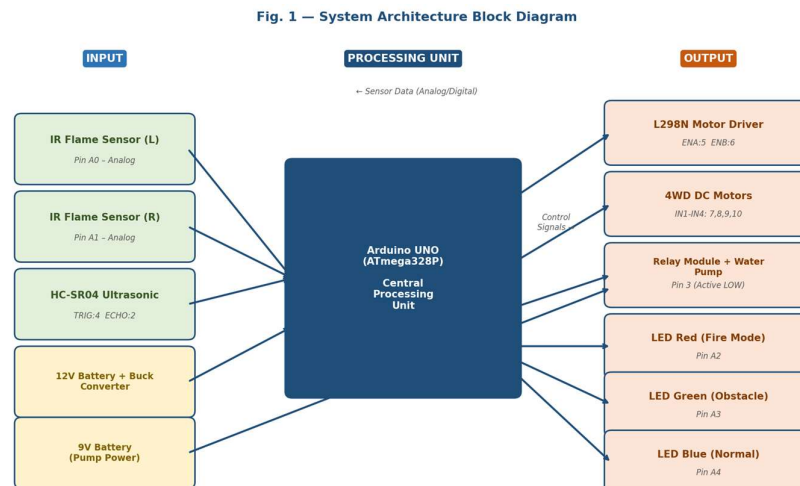


Fig. 1. System Architecture Block Diagram of the Fire Fighting and Obstacle Avoiding Robot

The Arduino UNO acts as the central processing unit, receiving analog signals from the dual IR flame sensors (A0, A1) and digital pulse signals from the HC-SR04 ultrasonic sensor (Pins 4, 2). It generates PWM and direction signals to the L298N motor driver, a relay control signal for the water pump (Pin 3), and digital signals to the three RGB LED state indicators (Pins A2–A4). A dual-rail power supply — 12V with Buck Converter for logic, 9V for pump — prevents voltage instability during pump motor startup.

B. Tinkercad Virtual Circuit Diagram

Fig. 2 shows the Tinkercad virtual circuit diagram illustrating the primary component connections and dual power rail configuration. This simulation was produced during the initial circuit design phase.

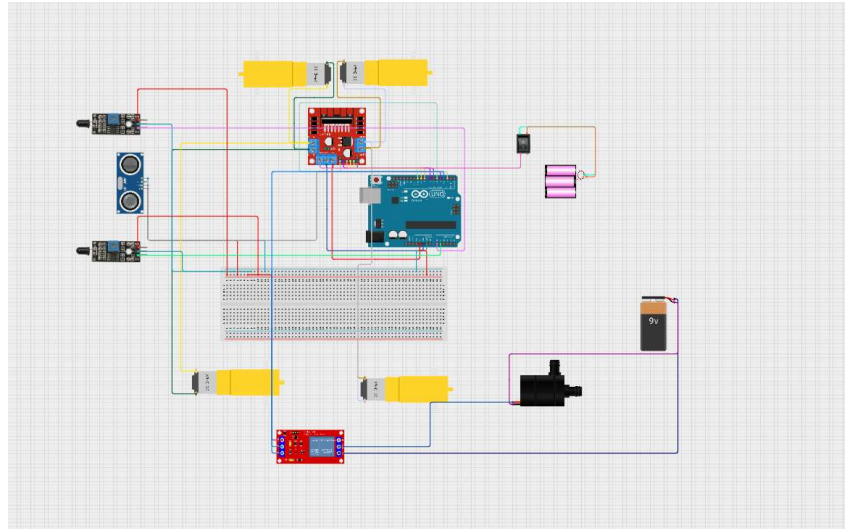


Fig. 2. Tinkercad Virtual Circuit Diagram — Fire Fighting and Obstacle Avoiding Robot

Diagram Note: RGB LED indicators (Pins A2–A4) and 12V Buck Converter were added to the physical prototype after this Tinkercad simulation was produced and are not yet reflected in the diagram above. These components are fully documented in Table I and are present in the physical build. Updated diagram pending from Shihad Md. Akash (22201203).

C. Hardware Components

1) Arduino UNO (ATmega328P) — Central Processing Unit

Operates at 5V logic with 14 digital I/O pins (6 PWM-capable) and 6 analog input pins. Reads all sensor data, executes the three-mode FSM, and outputs all control signals at a 50 ms loop cycle.

2) HC-SR04 Ultrasonic Sensor (TRIG: Pin 4, ECHO: Pin 2)

Emits 40 kHz pulses and measures echo time-of-flight to calculate distance. Accuracy: ± 3 mm at 20 cm [3]. Formula: $\text{distance (cm)} = \text{pulse duration (us)} / 58$. A 30 ms timeout prevents loop blocking on missing echoes. Obstacles within 20 cm trigger Obstacle Mode.

3) Dual IR Flame Sensors (FL_LEFT: A0, FL_RIGHT: A1)

Two flame sensor modules mounted left and right at the chassis front. Each detects IR radiation in the 760–1100 nm band. A 5-sample averaging filter (`detectFlame()`) eliminates false triggers from ambient IR. Analog reading below threshold 700 on either sensor triggers Fire Mode.

4) L298N Motor Driver (ENA: Pin 5, ENB: Pin 6, IN1–IN4: Pins 7–10)

Controls two DC motor pairs via PWM speed signals (ENA, ENB) and digital direction signals (IN1–IN4). Base speed: PWM 170 for forward; 150 for reverse and turns. Supports up to 2A per channel at 12V.

5) Mini Submersible Water Pump (Pin 3 via 5V Single Channel Relay)

Relay activated LOW on Pin 3 switches the pump's 9V power circuit. Dedicated 9V battery powers the pump independently to prevent voltage drops in the control circuit. Spray duration: sprayMs = 4000 ms. After extinguishing, robot reverses 500 ms to clear the area.

6) RGB LED State Indicators (LED_R: A2, LED_G: A3, LED_B: A4)

Three discrete LEDs provide real-time visual FSM state feedback: Red (A2) = Fire Mode; Green (A3) = Obstacle Mode; Blue (A4) = Normal navigation. Analog pins A2–A4 are used as digital outputs, preserving digital pins for the motor driver. Note: LEDs are present on the physical robot but not yet reflected in the Tinkercad diagram (Fig. 2); see Table I.

7) Dual-Rail Power Supply (12V Li-Ion + Buck Converter / 9V Battery)

12V Li-Ion battery pack with Buck Converter supplies Arduino and all logic. Separate 9V battery powers water pump exclusively. Dual-rail isolation was implemented after single-supply testing caused microcontroller resets due to pump motor inrush current. Note: Buck Converter present in physical build but not yet in Tinkercad diagram (Fig. 2).

D. Pin Assignment Summary

Component	Pin	Mode
L298N ENA — Left Motor Speed	Pin 5	PWM Output
L298N ENB — Right Motor Speed	Pin 6	PWM Output
Motor Direction IN1	Pin 7	Digital Output
Motor Direction IN2	Pin 8	Digital Output
Motor Direction IN3	Pin 9	Digital Output
Motor Direction IN4	Pin 10	Digital Output
HC-SR04 TRIG	Pin 4	Digital Output
HC-SR04 ECHO	Pin 2	Digital Input
Water Pump (via Relay Module)	Pin 3	Digital Output (Active LOW)
IR Flame Sensor Left	A0	Analog Input
IR Flame Sensor Right	A1	Analog Input
LED Red — Fire Mode	A2	Digital Output
LED Green — Obstacle Mode	A3	Digital Output
LED Blue — Normal Mode	A4	Digital Output

Table I. Complete Pin Assignment for Arduino UNO

IV. ALGORITHM

A. Control Strategy — Three-Mode Prioritized FSM

The robot implements a three-mode prioritized Finite-State Machine (FSM) in a continuous 50 ms loop. Fire Mode holds absolute priority; if flame is detected the robot stops and activates the pump regardless of obstacle status. Obstacle Mode is next; Normal Mode is the default when neither condition is active.

- Priority 1 — Fire Mode: Activated when detectFlame() returns TRUE. Robot halts immediately, activates pump for 4 s spray, then reverses 500 ms. LED: Red. All other modes suppressed while flame persists.
- Priority 2 — Obstacle Mode: Activated only when no flame detected and dist < 20 cm. Robot reverses 400 ms then pivots right 500 ms. LED: Green.
- Priority 3 — Normal Mode (Default): No flame, no obstacle. Robot moves forward at PWM 170. LED: Blue.

B. Algorithm Flowchart

Fig. 3 illustrates the complete three-mode FSM, including the continuous sensing loop, priority-ordered decision diamonds, mode-specific action blocks, and the 50 ms loop-back mechanism.

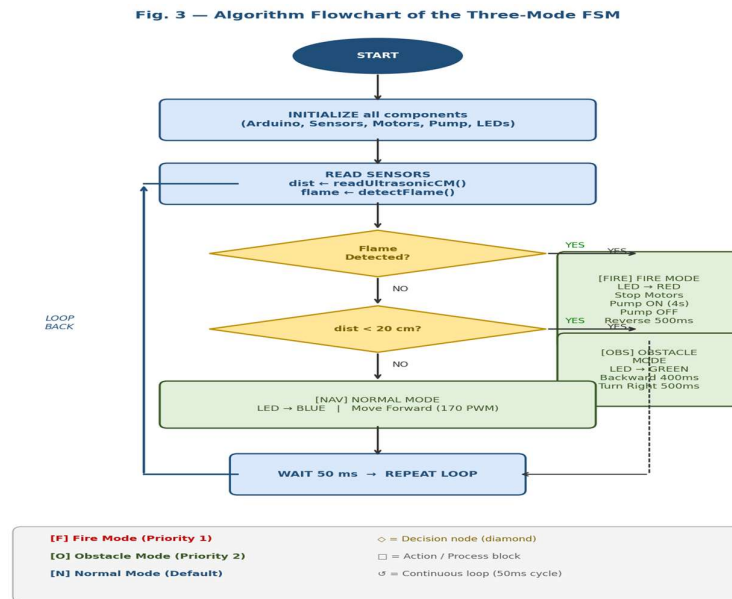


Fig. 3. Algorithm Flowchart of the Three-Mode Prioritized FSM (Fire > Obstacle > Normal)

C. Algorithm Pseudocode

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ALGORITHM: FireFightingAndObstacleAvoidingRobot


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PROCEDURE SETUP:
  Configure all GPIO pins as INPUT or OUTPUT
  SET PUMP = OFF ; Initialize Serial (9600 baud)

PROCEDURE LOOP:
  dist <- readUltrasonicCM()
  flame <- detectFlame()

  IF flame == TRUE THEN           // Fire Mode (Priority 1)
    SET LED <- RED
    CALL motorsStop() | WAIT 200 ms
    SET PUMP <- ON | WAIT 4000 ms
    SET PUMP <- OFF
    CALL motorsBackward(150,150) | WAIT 500 ms
    CALL motorsStop() | WAIT 300 ms

  ELSE IF dist < 20 cm THEN       // Obstacle Mode (Priority 2)
    SET LED <- GREEN
    CALL motorsBackward(150,150) | WAIT 400 ms
    CALL motorsTurnRight(150) | WAIT 500 ms

  ELSE                           // Normal Mode (Default)
    SET LED <- BLUE
    CALL motorsForward(170,170)
  END IF
  WAIT 50 ms | GOTO LOOP


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FUNCTION readUltrasonicCM() -> distance (cm):
  Pulse TRIG HIGH for 10 us
  dur <- pulseIn(ECHO, HIGH, timeout=30 ms)
  IF dur == 0 THEN RETURN 999
  RETURN dur / 58

FUNCTION detectFlame() -> Boolean:
  FOR i = 1 TO 5:
    sumL <- sumL + analogRead(FL_LEFT)
    sumR <- sumR + analogRead(FL_RIGHT)
  WAIT 5 ms
  END FOR
  RETURN (sumL/5 < flameThresh) OR (sumR/5 < flameThresh)


---



```

D. Key Sub-Algorithms

1) readUltrasonicCM()

TRIG pulsed HIGH for 10 us. pulseIn(ECHO, HIGH, 30000) measures echo width with 30 ms timeout. Returns 999 cm if no echo. Otherwise: distance = dur / 58, from speed of sound: round-trip (cm) = dur (us) x 0.0343 / 2.

2) detectFlame() — 5-Sample Averaging Filter

Reads both IR sensors 5 times with 5 ms inter-sample delay. Computes avgL = sumL/5 and avgR = sumR/5. Returns TRUE if either average falls below threshold 700. This filter eliminates single-

sample false positives from lighting flicker and electrical noise on the analog lines — critical for reliable fire detection.

V. DISCUSSION

A. Experimental Results

The assembled prototype was tested in a controlled indoor environment. Table II summarizes outcomes across the three primary test scenarios.

Test Condition	Robot Action Taken	Outcome
Obstacle at 10 cm	Reversed 400 ms + Pivoted Right 500 ms	Success
Flame within 25 cm	Halted + Pump ON 4 s + Reversed 500 ms	Success
No Obstacle / No Flame	Moved Forward (Blue LED lit)	Success
Flame + Obstacle simultaneously	Fire Mode activated (Priority 1 overrode Obstacle Mode)	Success

Table II. Experimental Performance Analysis

B. Strengths

The prioritized FSM is deterministic: fire suppression is never delayed by concurrent obstacle events, confirmed by simultaneous-stimulus testing. The dual-rail power supply eliminated the microcontroller reset failures observed in early single-supply testing. The 5-sample IR averaging filter reduced false-positive flame detections to zero in controlled indoor conditions. RGB LED indicators proved useful for real-time FSM state diagnostics during testing.

C. Limitations

The obstacle avoidance strategy is purely reactive and memoryless — always backing up and turning right, with no environment mapping. This can cause the robot to loop in concave obstacle layouts. The HC-SR04 has a narrow (~15 degree) detection cone, creating lateral blind spots. IR flame sensors require threshold recalibration per environment and are susceptible to high-ambient-IR sources. The Tinkercad diagram (Fig. 2) does not yet reflect the RGB LED indicators and Buck Converter; an updated version is pending from Shihad Md. Akash.

D. Future Improvements

Planned Phase 2 enhancements include: (1) servo-mounted HC-SR04 for a wider angular obstacle scan; (2) wall-following or Pledge algorithm for complex environments; (3) ESP32-CAM for camera-based flame detection; (4) ESP32 IoT module for remote monitoring; (5) DHT11 temperature/humidity sensor for environmental logging.

VI. CONCLUSION

This project successfully demonstrates the design and implementation of a low-cost autonomous Fire Fighting System and Obstacle Avoiding Robot. By integrating an Arduino UNO microcontroller with an HC-SR04 ultrasonic sensor, dual IR flame sensors, an L298N motor driver, three RGB LED state indicators, and a relay-controlled submersible water pump on a 4WD acrylic chassis, the system achieves reliable autonomous operation within a prioritized three-mode FSM.

Experimental results confirm that the robot correctly suppresses fires upon detecting flames within 25 cm, avoids obstacles within 20 cm, and accurately reflects its FSM state via LED feedback at all times. The 5-sample IR averaging filter proved essential for eliminating false-positive fire events. The dual-rail power supply resolved microcontroller instability caused by pump inrush current. The total active hardware cost of approximately BDT 3,800 confirms that effective embedded robotic safety systems are achievable at low cost — making deployment feasible in garment factories, manufacturing units, and other fire-prone environments across Bangladesh.

Future work will focus on camera-based fire detection, IoT remote monitoring, servo-mounted ultrasonic scanning, and more sophisticated navigation algorithms to extend system capability in complex real-world environments.

CONTRIBUTIONS OF EACH MEMBER

Name	Student ID	Email	Contributions
Md. Shahriar Hossain	22201196	22201196@uap-bd.edu	Survey 1 (Market Survey), Survey 2 (Impact Assessment), System Definition and three-mode logic concept, Report writing — Introduction, Related Works, Discussion, Survey Appendices A and B
Roudro Saha	22201205	22201205@uap-bd.edu	Algorithm design and pseudocode, Arduino code implementation — FSM loop logic, detectFlame() averaging filter, all motor control and actuator control functions
Shihad Md. Akash	22201203	22201203@uap-bd.edu	Physical robot construction and hardware assembly, component mounting on chassis, circuit wiring, Tinkercad virtual circuit diagram (update pending to add RGB LEDs and Buck Converter)
Abdullah al Masud	22201200	22201200@uap-bd.edu	Hardware setup assistance and component integration, survey activities support, system-level testing and experimental results documentation

Appendix A. SURVEY 1 — MARKET SURVEY

Conducted by: Md. Shahriar Hossain (22201196)

A.1 Objective

To identify the most cost-effective and technically suitable electronic components available in the Bangladeshi electronics market for a fire-fighting and obstacle-avoiding robot, with specific emphasis on flame-detection sensors appropriate for high-heat industrial environments.

A.2 Methodology

The market survey was conducted through online inquiry and reference to local electronics retailers in Dhaka (primarily Elephant Road electronics market and online platforms Techshopbd and Robu.in.bd). Component availability, unit pricing, electrical specifications, and Arduino UNO compatibility were evaluated for each candidate type.

A.3 Flame Sensor Comparison

Sensor Type	Price (BDT)	Range	Arduino Compatible	Selected
IR Flame Sensor (KY-026)	150	25–100 cm	Yes (Analog + Digital)	YES
UV Flame Sensor	450–600	50–200 cm	Yes (Digital only)	No
Thermocouple (Type K)	500–800	Contact only	Requires amplifier	No
DHT11 (Temp/Humidity)	120	Ambient only	Yes (no flame detection)	No

Table A1. Flame Sensor Market Comparison (Dhaka, Bangladesh)

The IR Flame Sensor Module (KY-026) was selected for cost-effectiveness (BDT 150/unit), direct analog output enabling the Arduino ADC noise-averaging algorithm, and sensitivity to the 760–1100 nm infrared band emitted by open flames in high-heat environments.

A.4 Component Procurement Budget

Component	Qty	Unit Price (BDT)	Total (BDT)
Arduino UNO (Clone)	1	550	550
L298N Motor Driver	1	300	300
DC Geared Motors	4	200	800
4WD Robot Chassis (with wheels)	1 set	180	180
Water Pump (submersible)	1	400	400
Water Container + Pipe	1 set	150	150
IR Flame Sensor Module	3	180	540
HC-SR04 Ultrasonic Sensor	1	180	180
Li-Ion Battery Pack (18650)	1	500	500
5V Single Channel Relay Module	1	100	100
BC547 Transistor + Resistors	—	30	30
Misc. (Wires, Breadboard, LEDs)	—	170	170
ACTIVE COMPONENT TOTAL			3,800
Servo Motor ** (Phase 2 Reserve)	1	350	350
FULL PROCUREMENT TOTAL			4,150

Table A2. Component Procurement Budget

** Servo Motor (BDT 350) was procured as a Phase 2 component for the planned servo-mounted ultrasonic sensor upgrade (see Section V.D). It is not part of the active FSM implementation and is excluded from the active component total of BDT 3,800.

Appendix B. SURVEY 2 — IMPACT ASSESSMENT

Conducted by: Md. Shahriar Hossain (22201196)

B.1 Objective

To assess the real-world social and safety impact of deploying the Fire Fighting and Obstacle Avoiding Robot in high-risk industrial and residential environments in Bangladesh, and to justify the target deployment sector.

B.2 High-Risk Sectors Identified

- Garment and Ready-Made Garment (RMG) Sector: Bangladesh is the world's second-largest garment exporter, with densely packed factory floors containing flammable fabrics and limited evacuation paths. Historical incidents (Tazreen Fashion Fire 2012: 112 deaths; Nimtoli Fire 2010: 123 deaths) underscore the acute need for autonomous early fire response systems.
- Manufacturing and Industrial Plants: BSCIC and BEPZA export processing zones face fire risks from flammable raw materials. An autonomous firefighting robot can suppress early-stage fires before escalation, reducing casualties and equipment losses.

- Residential High-Rise Buildings: Rapid high-rise construction in Dhaka, Chittagong, and Sylhet creates fire safety gaps where limited fire-service response time leaves occupants vulnerable. Compact robots can supplement building fire safety systems.

B.3 Impact Summary

Impact Dimension	Assessment
Human Risk Reduction	High — robot removes direct human presence from the fire zone during early response
Deployment Cost (active)	Low (BDT 3,800/unit) — scalable for SMEs and garment factories across Bangladesh
Response Latency	Near-instantaneous (< 50 ms FSM loop; no human decision delay)
False-Positive Rate	Low in indoor environments with 5-sample IR averaging filter active
Scalability	High — modular design supports ESP32-CAM, IoT, servo, DHT upgrades in Phase 2
STEM Education Value	High — demonstrates embedded FSM design and sensor integration in CSE 438
IoT Upgrade Potential	High — ESP32 addition enables real-time remote monitoring and SMS/alert systems

Table B1. Social and Technical Impact Summary

Appendix C. FULL ARDUINO SOURCE CODE

Compiled and uploaded using Arduino IDE 2.x targeting Arduino UNO (ATmega328P). Code authored by Roudro Saha (22201205). **Note: #include <Servo.h> has been intentionally excluded — the servo motor is a Phase 2 component not used in the current FSM.**

```
// Fire Fighting System and Obstacle Avoiding Robot
// CSE 438 - Robotics Lab, University of Asia Pacific
// Platform: Arduino UNO (ATmega328P)
// Author: Roudro Saha (22201205)
// Note: Servo.h intentionally excluded - servo is a Phase 2 component.

// ---- Pin Map ----
const int ENA=5, IN1=7, IN2=8, IN3=9, IN4=10, ENB=6;
const int TRIG=4, ECHO=2;
const int FL_LEFT=A0, FL_RIGHT=A1;
const int PUMP=3, LED_R=A2, LED_G=A3, LED_B=A4;

// ---- Tunables ----
int baseSpeed=170, turnSpeed=150, avoidDist=20;
int flameThresh=700, sprayMs=4000;

// ---- Motor Control ----
void motorsStop() { analogWrite(ENA,0); analogWrite(ENB,0); }

void motorsForward(int sL,int sR) {
  digitalWrite(IN1,HIGH); digitalWrite(IN2,LOW);
  digitalWrite(IN3,HIGH); digitalWrite(IN4,LOW);
  analogWrite(ENA,sL);   analogWrite(ENB,sR); }

void motorsBackward(int sL,int sR) {
```

```

digitalWrite(IN1,LOW); digitalWrite(IN2,HIGH);
digitalWrite(IN3,LOW); digitalWrite(IN4,HIGH);
analogWrite(ENA,sL);   analogWrite(ENB,sR); }

void motorsTurnRight(int s) {
  digitalWrite(IN1,HIGH); digitalWrite(IN2,LOW);
  digitalWrite(IN3,LOW); digitalWrite(IN4,HIGH);
  analogWrite(ENA,s);   analogWrite(ENB,s); }

// ---- Ultrasonic Distance ----
long readUltrasonicCM() {
  digitalWrite(TRIG,LOW); delayMicroseconds(2);
  digitalWrite(TRIG,HIGH); delayMicroseconds(10);
  digitalWrite(TRIG,LOW);
  long dur = pulseIn(ECHO,HIGH,30000);
  return (dur==0) ? 999 : dur/58; }

// ---- Actuator Helpers ----
void pumpOn(bool on) { digitalWrite(PUMP, on ? LOW : HIGH); }
void setLED(bool r,bool g,bool b) {
  digitalWrite(LED_R,r); digitalWrite(LED_G,g); digitalWrite(LED_B,b); }

// ---- Averaged Flame Detection (5-sample noise filter) ----
bool detectFlame() {
  int sumL=0, sumR=0;
  for (int i=0; i<5; i++) {
    sumL += analogRead(FL_LEFT);
    sumR += analogRead(FL_RIGHT);
    delay(5); }
  return (sumL/5 < flameThresh || sumR/5 < flameThresh); }

// ---- Setup ----
void setup() {
  Serial.begin(9600);
  int outs[]={ENA,ENB,IN1,IN2,IN3,IN4,PUMP,LED_R,LED_G,LED_B};
  for (int p:outs) pinMode(p,OUTPUT);
  pinMode(TRIG,OUTPUT); pinMode(ECHO,INPUT);
  pumpOn(false); }

// ---- Main Loop (Three-Mode FSM) ----
void loop() {
  long dist = readUltrasonicCM();
  bool flame = detectFlame();
  Serial.print("Dist: "); Serial.print(dist);
  Serial.print(" cm | Flame: "); Serial.println(flame);

  // Priority 1: FIRE MODE
  if (flame) {
    setLED(true,false,false); // Red LED
    motorsStop(); delay(200);
    pumpOn(true); delay(sprayMs);
    pumpOn(false);
    motorsBackward(150,150); delay(500);
    motorsStop(); delay(300);

    // Priority 2: OBSTACLE MODE
  } else if (dist < avoidDist) {
    setLED(false,true,false); // Green LED
    motorsBackward(150,150); delay(400);
    motorsTurnRight(turnSpeed); delay(500);

    // Priority 3: NORMAL MODE
  } else {
    setLED(false,false,true); // Blue LED
    motorsForward(baseSpeed,baseSpeed);
  }
  delay(50);
}

```

REFERENCES

- [1] K. Pathak, B. B. Sah and A. Pandey, "Design and Development of an Ultrasonic Obstacle Avoiding Robot," *International Journal of Engineering Research & Technology*, vol. 6, no. 5, pp. 123-127, 2017.
- [2] M. A. Hossain and M. M. Rahman, "Obstacle Avoidance Robot using Arduino and Ultrasonic Sensor," *American Journal of Engineering Research*, vol. 5, no. 2, pp. 64-69, 2016.
- [3] N. S. Kumar, S. S. Kumar and R. P. Raj, "Autonomous Robot for Obstacle Detection using HC-SR04 Sensor," in *Proc. ICIIECS*, IEEE, 2015.
- [4] A. Tiwari and R. Kumar, "Fire Fighting Robot Controlled Using Sensors," *International Journal of Science and Research*, vol. 4, no. 4, pp. 234-238, 2015.
- [5] D. S. Kumbhar and R. Patil, "Development of Small-Scale Fire-Fighting Robot," in *Proc. ICCICCT*, IEEE, 2014.
- [6] D. Mishra, "Design and Implementation of a Fire Detection and Extinguishing Robot," *International Journal of Advanced Research in Electrical, Electronics and Instrumentation Engineering*, vol. 5, no. 3, pp. 188-195, 2016.
- [7] R. V. Kulkarni and A. S. Sali, "Sensor Based Automation System Using Arduino," in *Proc. CAST*, IEEE, 2016.
- [8] J. G. Mendoza, "Integration of Multiple Sensors for Robot Navigation," *IEEE Transactions on Industrial Electronics*, vol. 58, no. 8, pp. 3395-3404, 2011.
- [9] G. Dudek and M. Jenkin, *Computational Principles of Mobile Robotics*, 2nd ed., Cambridge University Press, 2010.
- [10] B. Siciliano and O. Khatib, *Springer Handbook of Robotics*. Springer, 2016.
- [11] J. J. Craig, *Introduction to Robotics: Mechanics and Control*, 3rd ed., Pearson, 2004.
- [12] S. K. Saha, "Autonomous Mobile Robotics for Industrial Applications," in *Proc. IEEE CASE*, 2018.
- [13] M. R. Islam and A. M. N. Islam, "Design of a Fire Detection and Suppression Robot for Household Safety," in *Proc. International Conference on Engineering Research and Education*, 2020.
- [14] P. Singh, "IoT-Based Fire Detection and Control System," in *Proc. IEEE International Conference on Computing, Communication and Automation*, 2018.